

SMART INDIA HACKATHON 2026

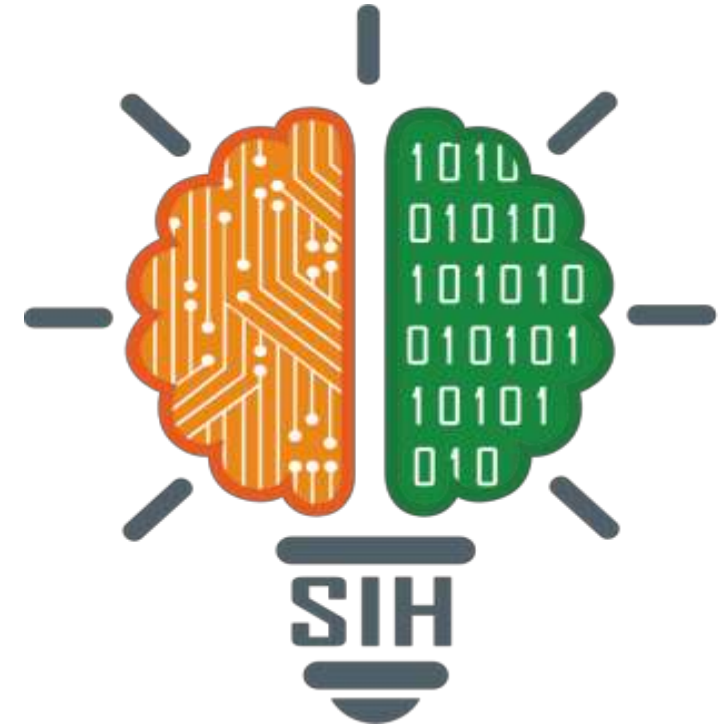
TITLE PAGE



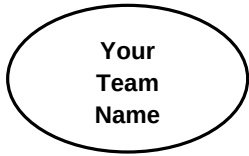
AI-Based Smart Fertilizer Recommendation System

Precision Nutrient Optimization & Decision-Support Engine for Indian Agriculture

- Problem Statement ID – [INSERT ACTUAL PS ID]
- Theme – Agriculture, FoodTech & Rural Development
- Team Name – [INSERT REGISTERED TEAM NAME]



SOIL CHEMISTRY (N,P,K,pH,Micros) → ML SOFT-VOTING ENSEMBLE → PRECISE FERTILIZER DOSAGE → BETTER CROP YIELD



IDEA TITLE



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

■ THE PROBLEM

- Farmers apply fertilizers without knowing actual soil nutrient status (guesswork).
- Incorrect choice or dosage inflates input costs and reduces farmer profit margins.
- Excessive chemicals cause long-term soil acidification & water eutrophication.
- Manual interpretation of 12-parameter soil health cards is complex for farmers.

■ PROPOSED SOLUTION

- AI/ML decision-support system analyzing 12+ soil parameters (N, P, K, pH, Micros).
- Crop-specific intelligence aligning with exact crop stoichiometric requirements.
- Recommends optimum commercial fertilizer product + precise application dosage (kg/ha).
- Generates simple, actionable advice with transparent soil justification reports.

■ INNOVATION & UNIQUENESS

- Integrates Liebig's Law of the Minimum across all 6 essential trace micronutrients.
- Soft-Voting Ensemble combining Trees (RF, ET), HistGradientBoosting & MLP.
- Validated against empirical distributions of 10.85M National Soil Database records.
- Interactive web platform with weather-safe application and spray guidance.

SYSTEM PIPELINE: [Soil Data + Crop] → [Data Preprocessing & Scaling] → [Trained ML Model] → [Fertilizer Type + Dosage] → [Farmer Report]



■ END-TO-END SYSTEM ARCHITECTURE

1. INPUT INGESTION: Soil N, P, K, pH, OC, EC, Zn, B, S, Fe, Mn, Cu + Crop Choice
2. WEB UI LAYER: Responsive Django portal with field-level input sanitization
3. PREPROCESSING: Outlier filtering & RobustScaler fit strictly on training partition
4. FEATURE ENGINEERING: 44 stoichiometric ratios, Liebig factors & soil buffers
5. ML META-ENSEMBLE: Soft-Voting (Random Forest, Extra Trees, HGB, Deep MLP)
6. STOICHIOMETRIC ENGINE: Exact stoichiometric balance calculation for kg/ha dosage
7. EXPLAINABLE DASHBOARD: Scientific justification drivers & weather spray warnings

■ TECHNOLOGY STACK

- Frontend: HTML5, Modern Responsive CSS, Vanilla JS
- Backend: Django Framework, Python 3.14 REST API
- Machine Learning: Scikit-learn, NumPy, Pandas, Joblib
- Model Suite: RF (250) + Extra Trees (250) + HGB + MLP
- Database: SQLite (Indexed National Soil DB + Farm Plots)

■ ML LIFECYCLE & WORKING PROTOTYPE

- ML Lifecycle: National Soil Data (10.85M) → 5-Fold Stratified Cross-Validation → Final Holdout Test → Live API Deployment
- Working Prototype: Full functional Django web application delivering real-time predictions (<50ms) with PDF export.

FEASIBILITY AND VIABILITY

■ ■ TECHNICAL FEASIBILITY

- Mature Scikit-learn & Python stack ensures production grade stability.
- Django enables rapid, modular, and secure web deployment.
- Lightweight model artifacts (<400MB) enable instant real-time inference.
- Easily connects to IoT soil sensors and meteorological live APIs.

■ OPERATIONAL FEASIBILITY

- Clean 2-column input UI designed for extension officers and village centers.
- Outputs delivered in standard local farmer metrics (kg/ha and commercial bags).
- Explainable justifications build strong confidence among farmers.
- Operates on basic mobile browsers with low bandwidth requirements.

■ ■ CHALLENGES & MITIGATIONS

- Soil Data Quality Variance:
Mitigation: Automated range sanitization & RobustScaler handling.
- Model Reliability Risks:
Mitigation: 5-Fold cross-validation & abstention flags for OOD inputs.
- Farmer Trust & Adoption:
Mitigation: Transparent, explainable rules.
- Regional Soil Diversity:
Mitigation: Empirical calibration across 32 Indian states and 735 districts.

FUTURE SCALABILITY: [Web Portal Prototype] → [Multilingual Mobile App] → [Govt SHC Database API] → [IoT Soil Sensor Grid]



SOIL HEALTH DATA → INTELLIGENT AI RECOMMENDATION → SUSTAINABLE FARM OUTCOMES

■■■ FARMERS

- Eliminates trial-and-error fertilizer purchasing.
- Tailored precisely to crop nutrient absorption rates.
- Easy-to-follow advice in practical farmer units.

■ ECONOMIC

- Reduces unnecessary input costs on surplus fertilizer.
- Enhances crop yield efficiency and ROI.
- Prevents crop lodging from nitrogen overdose.

■ ENVIRONMENTAL

- Curbs nitrate leaching into groundwater aquifers.
- Protects soil microbiomes and prevents acidification.
- Supports long-term national soil conservation.

■ TECH & SOCIAL

- Democratizes scientific precision agronomy.
- Empowers agricultural extension field staff.
- Scalable architecture ready for rural digital grids.

FUTURE POTENTIAL: Regional Agro-Ecological Models | IoT Soil Probes | Live Weather Feeds | Multilingual Voice Bot



■ AGRONOMIC & TECHNICAL REFERENCES

- ICAR Guidelines: Soil Testing & Fertilizer Recommendation Handbook (Indian Council of Agricultural Research).
- National Soil Health Card Database: 10,853,209 survey records analyzed across 735 districts.
- Liebig's Law of the Minimum: Multi-nutrient limiting factor formulation across 6 essential micronutrients.
- Scikit-learn & Python Docs: Voting Classifier, RobustScaler, and Stratified K-Fold validation architectures.
- Django Framework Documentation: High-performance RESTful web API and model serving.

■ MEASURED ML VALIDATION METRICS

- Holdout Test Accuracy: 99.75% (14,907 / 14,945 strictly unseen test samples)
- Macro F1-Score: 0.9961 | Weighted F1: 0.9975
- Macro F3-Score (beta=3.0): 0.9960 | Weighted F3: 0.9975
- Top-2 Accuracy: 100.00% | Top-3 Accuracy: 100.00%
- Multi-Class Log Loss: 0.0680 | Macro Brier Score: 0.0019
- Regional Empirical Validation: 10.85M records across 32 States
- Zero-Data-Leakage Preprocessing: Scaler fitted on Train ONLY
- Scientific Validation Status: PARTIALLY VALIDATED
(Technical & Synthetic Rules: PASSED | Multi-Year Farm Field Trials: Required)

PROJECT LINKS: [GitHub: Shivam-Patel22/SIH-2026-Project-] | [Working Demo: /api/recommendations/generate/] | [Audit: model_audit.json]